



Fig. 9: RAM missile being launched and radar elements of SSDS (inset)

Though this system is an automated C² system, it has a manual override feature. For the commander, SSDS suggests the mode of engagement for incoming targets. In manual mode, it leaves the responsibility of engagement to the operator. In automated mode, it takes care of the engagement itself. This automation reduces the time taken from detection to engagement.

AN/SPS-49 long-range air-search radar. This radar can detect incoming cruise missiles and aircraft flying very high but has restrictions in detecting low flyers.

AN/SPS-67 surface-search/navigation radar. This is a 2D radar that operates in 5450MHz to 5825 MHz region (G band). It is used for surface surveillance and navigation. This capability of the radar results in it having a secondary capability of detecting low-flying cruise missiles. The radar's detection range is around 100km and it is capable of a peak power of 280kW.

It is flexible in a way that pulse widths and pulse-repetition frequencies (PRF) of radar energy can be varied, accordingly, for the desired resolution. A higher PRF setting and short pulse width increases the resolution to detect two smaller targets in close proximity to each other. Other settings may even show these two targets as a single large target. But other settings also have different

uses; the operator chooses the settings accordingly.

AN/SPQ-9 air and surface surveillance radar. This is an air and surface surveillance X band-phased array radar. It covers 360° in azimuth at the rate of 30rpm. This allows the radar to detect sea-skimming missiles at the horizon

against the presence of heavy sea clutter. It automatically displays any target with a relative speed exceeding ten knots. It complements high-altitude surveillance radars by detecting missiles that are approaching just above sea surface.

In surface mode, it generates a separate surface frequency to detect surface targets. It can even detect a submarine that has partially submerged and is just peeking its periscope above the water surface.

The radar is present in Aegis vessels also, where it helps these to fire accurately on the enemy surface vessels.

Rolling airframe missile (RAM).

This missile is carried in bulk in its launcher. It is capable of reaching mach 2 speeds with an effective range of 9km. It is independent of any radar and has a passive radar-homing guidance. It targets the incoming cruise missile by homing on its radar emissions.

Generally, anti-ship cruise missiles have radars as seekers. RAM looks for radar emissions, finds these, locks on these and flies towards these. Also, RAM can target infrared (IR) emissions that are abundantly available in the incoming cruise missile's engine exhaust. As the missile's exhaust is very hot, it emits IR radiations like a hot iron emitting red light.

Both Sea Sparrow and RAM have a minimum range beyond only which

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